Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_27

1. Summary of Key Deviations

The analysis of Run_27 reveals a critical and highly isolated sensor failure, with the rest of the system showing only minor, likely secondary, deviations.

- **p_FMU (Critical):** This is the primary and most severe anomaly. The pressure sensor at the Fuel Metering Unit shows a **major negative drift** starting in state s2, which worsens until it becomes a **complete signal dropout (-99.4%)** in state s5. The signal partially recovers but remains at a major negative offset (-68%) for the rest of the run.
- **p_Combustion (Minor):** A consistent, minor **negative offset** of approximately **-5% to -6%** is observed across all active states.
- **Flow_Bypass & Flow_Pump (Minor):** Both of these flow signals show a slight **positive offset** (around +8% and +5% respectively).
- **All Other Signals (Nominal):** Crucially, all four `Flow_Engine` signals remain perfectly healthy throughout the run, showing no significant deviation. `Flow_PRV` and all other pressure signals are also nominal.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**p_FMU Sensor**	**Sensor Dropout / Total Failure**	This is the most probable cause. The `p_FMU` sensor signal drops to nearly -100% and stays there, which is a classic dropout failure. Critically, this happens in complete isolation from any expected physical consequences. A real, total loss of pressure at the FMU would starve the engines of fuel, causing all `Flow_Engine` signals to drop to zero. Instead, they remain perfectly healthy. This proves the physical system is fine and only the sensor has failed.

2	**Wiring / Electrical Fault to p_FMU**	**Signal Loss / Grounding Issue**	This is physically equivalent to a sensor failure. A break in the wiring or a grounding fault in the `p_FMU` sensor's circuit would result in a lost signal, explaining the dropout reading without any actual change in system pressure.
3	**Fuel Metering Unit (FMU)**	**Major Internal Leak**	A massive leak *within* the FMU could theoretically cause a localized pressure drop at that specific sensor. However, such a large leak would divert flow and almost certainly affect the downstream `Flow_Engine` readings, which are perfectly healthy. Therefore, a physical leak is highly unlikely.
4	**p_Combustion Sensor**	**Minor Calibration Shift**	The minor negative offset on `p_Combustion` could be a separate, small calibration error on that sensor. It is not significant enough to be the primary fault and is unrelated to the critical dropout of `p_FMU`.
5	**System Contamination**	**Debris affecting p_FMU sensor port**	It is conceivable that a piece of debris could have blocked the sensing port for the `p_FMU` sensor, causing it to read incorrectly. This is a possible underlying cause for the sensor failure.

3. Most Probable Hypothesis

The most probable root cause is a **Total Sensor Failure of the p_FMU Sensor**, likely due to an internal fault or an electrical/wiring issue.

Reasoning: The evidence for an isolated sensor failure is definitive. The `p_FMU` reading drops to nearly zero, but this has **no effect on the downstream engine flows**. The `Flow_Engine1-4` signals all remain nominal, which would be physically impossible if the pressure at the FMU had actually vanished. The system continues to operate as if nothing is wrong, proving that the pressure is still there and only the measurement is lost. The other minor flow and pressure deviations are likely unrelated sensor drift or noise, completely overshadowed by the critical dropout of the `p_FMU` sensor.